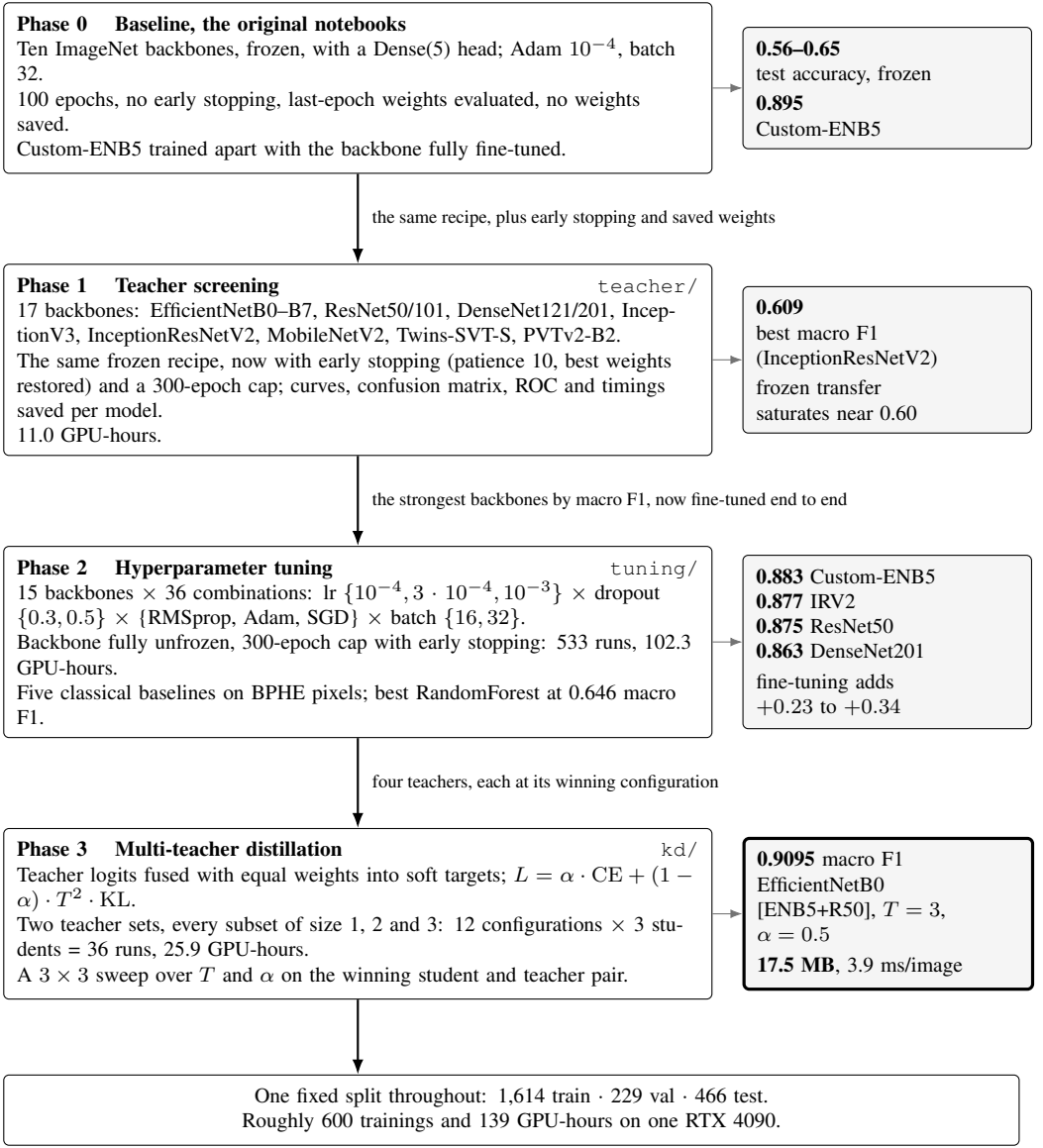
Phase 0 Baseline, the original notebooks
Ten ImageNet backbones, frozen, with a Dense(5) head; Adam 10^{-4}, batch 32.
100 epochs, no early stopping, last-epoch weights evaluated, no weights saved.
Custom-ENB5 trained apart with the backbone fully fine-tuned.
0.56–0.65
test accuracy, frozen
0.895
Custom-ENB5
the same recipe, plus early stopping and saved weights
Phase 1 Teacher screening
teacher/
17 backbones: EfficientNetB0–B7, ResNet50/101, DenseNet121/201, InceptionV3, InceptionResNetV2, MobileNetV2, Twins-SVT-S, PVTv2-B2.
The same frozen recipe, now with early stopping (patience 10, best weights restored) and a 300-epoch cap; curves, confusion matrix, ROC and timings saved per model.
11.0 GPU-hours.
0.609
best macro F1
(InceptionResNetV2)
frozen transfer
saturates near 0.60
the strongest backbones by macro F1, now fine-tuned end to end
Phase 2 Hyperparameter tuning
tuning/
15 backbones × 36 combinations: lr $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}\}$ × dropout $\{0.3, 0.5\}$ × {RMSprop, Adam, SGD} × batch $\{16, 32\}$.
Backbone fully unfrozen, 300-epoch cap with early stopping: 533 runs, 102.3 GPU-hours.
Five classical baselines on BPHE pixels; best RandomForest at 0.646 macro F1.
0.883 Custom-ENB5
0.877 IRV2
0.875 ResNet50
0.863 DenseNet201
fine-tuning adds
+0.23 to +0.34
four teachers, each at its winning configuration
Phase 3 Multi-teacher distillation
kd/
Teacher logits fused with equal weights into soft targets; $L = \alpha \cdot \mathrm{CE} + (1 - \alpha) \cdot T^2 \cdot \mathrm{KL}$.
Two teacher sets, every subset of size 1, 2 and 3: 12 configurations × 3 students = 36 runs, 25.9 GPU-hours.
A 3 × 3 sweep over T and α on the winning student and teacher pair.
0.9095 macro F1
EfficientNetB0
[ENB5+R50], $T = 3$,
$\alpha = 0.5$
17.5 MB, 3.9 ms/image
One fixed split throughout: 1,614 train · 229 val · 466 test.
Roughly 600 trainings and 139 GPU-hours on one RTX 4090.